P R a V A H

Predictive Routing and Vehicle’s Augmented Handling

Team SATVA

Details of Team Members

Sr.NO	Name	Degree Program	Year	Institution
1	Tanishka Atul Doiphode	B-tech	2nd year	Indian Institute of Technology Indore
2	Vedansh Shrivastava	B-tech	3rd year	Indian Institute of Technology Indore
3	Anuvab Mandal	B-tech	3rd year	Indian Institute of Technology Indore
4	Ayushi Vishwakarma	B-tech	2nd year	Indian Institute of Technology Indore
5	Gujjula Siri Sahasra	B-tech	2nd year	Indian Institute of Technology Indore

The Problem

Static traffic lights cannot adapt to ever-changing urban traffic density or road conditions. Thus, urban areas suffer from severe traffic unrest:

- congestions;
- Inefficient signal timings;
- Slow emergency response;
- Waste of time and fuel;
- Higher accident and pollution rates;

Size of the market

- India has over a million signalised intersections, most of which use fixed-time systems.
- The global smart traffic management market was valued at approximately $12 billion in 2023 and is projected to reach over $ 20 billion by 2030, with a compound annual growth rate (CAGR) of around 8–9%.
- Urban mobility solutions are a core focus under Smart Cities Mission and AI dreiven solutions, with 100+ Indian cities actively adopting smart infrastructure.
- Immediate target market: Urban and semi-urban municipalities, traffic enforcement agencies, and emergency response departments.
- A better, more reliable, and cost-efficient solution compared to existing or already implemented solutions.

Top 10 Indian cities with slowest traffic in 2024-25

Rank	City	State	Average Travel Time per 10 km	Time lost per year during rush hours	Global Rank
1	Kolkata	West Bengal	34 minutes 33 seconds	110 hours	2
2	Bengaluru	Karnataka	34 minutes 10 seconds	117 hours	3
3	Pune	Maharashtra	33 minutes 22 seconds	108 hours	4
4	Hyderabad	Telangana	31 minutes 30 seconds	85 hours	18
5	Chennai	Tamil Nadu	30 minutes 20 seconds	94 hours	31
6	Mumbai	Maharashtra	29 minutes 26 seconds	103 hours	39
7	Ahmedabad	Gujarat	29 minutes 3 seconds	73 hours	43
8	Ernakulam	Kerala	28 minutes 30 seconds	88 hours	50
9	Jaipur	Rajasthan	28 minutes 28 seconds	83 hours	52
10	New Delhi	Delhi	23 minutes 24 seconds	76 hours	122

Source: *Tom Tom Traffic Index 2024*

Note: *The TomTom Traffic Index ranked these cities based on average travel times over a standardized 10-kilometer distance, providing a consistent measure of congestion across diverse urban areas.*

Target Market & Opportunity

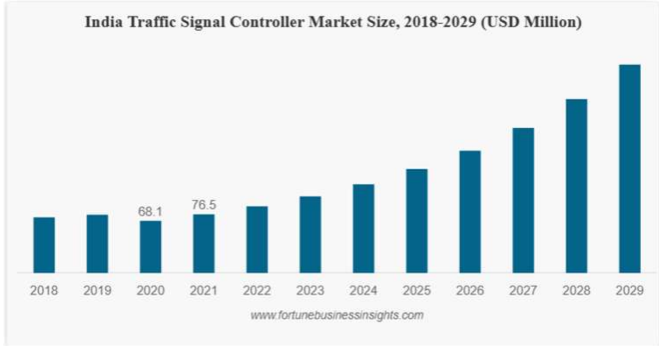
User Segments

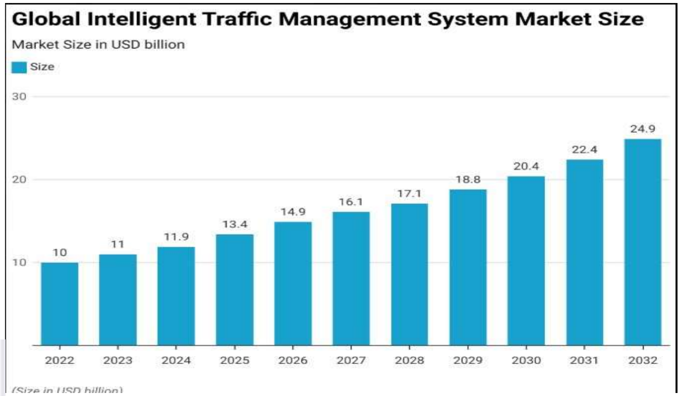
- **Urban & Semi-Urban Municipalities** - upgrading to adaptive AI traffic control for congestion reduction and commuter efficiency.
- **Traffic Police & Control Centres** - need real-time data dashboards and automated alerts for better decision-making.
- **Emergency Services** - ambulances, fire brigades, police vehicles needing prioritised routes.
- **Smart City Mission Planners** - integrating PRaVAH as part of broader IoT-driven infrastructure.
- **Highway Authorities / Toll Management** - for traffic flow optimisation and accident prevention.
- **Geographical Regions -** For traffic signals, where environmental conditions evoke low visibility.

Market Size & Opportunity

India Traffic Signal Controller Market:

- Valued at **USD 76.5 million in 2021**, projected to reach **USD 271.8 million by 2029** (CAGR 17.7%) - *Fortune Business Insights (2025). See ref.*
- Smart (AI/IoT-based) controllers form the **fastest-growing segment**, driven by Smart City investments (USD 27.6 billion allocated).
- Current AI-based VAC signals cost around **₹40–50 lakh per junction** (Bengaluru's ₹53 crore for 125 signals → ₹42.4 lakh each).
  In contrast, **PRaVAH delivers similar adaptive intelligence prototype at just ₹30,000 per junction => 99% cost reduction**.





Source: *Data Bridge Market Research, "Global Intelligent Traffic Management System Market Report, 2024–2032"*

Design: PRaVAH prototype

We have successfully designed and modelled the core of our system: the **PRaVAH edge device**. This is a field-ready, intelligent sensor hub engineered for real-world deployment on traffic poles.

Key Features of Our Prototype:

- **Multi-Sensor Array:** The unit houses **dual cameras to use the stereoscopic effect for redundant depth perception, along with a dedicated millimetre-wave radar** for reliable performance in low-visibility conditions like fog and heavy rain.
- **Onboard Intelligence:** An integrated SBC **(Raspberry Pi with USB AI accelerator)** performs all data processing and runs our AI models directly at the intersection.
- **Robust, All-Weather Design:** The device is built for Indian conditions, featuring a durable casing, a protective transparent shield for the cameras, and **heat-dissipating vents.**
- **Simple & Secure Mounting:** We've designed a practical mounting system using standard **U-bolts.**

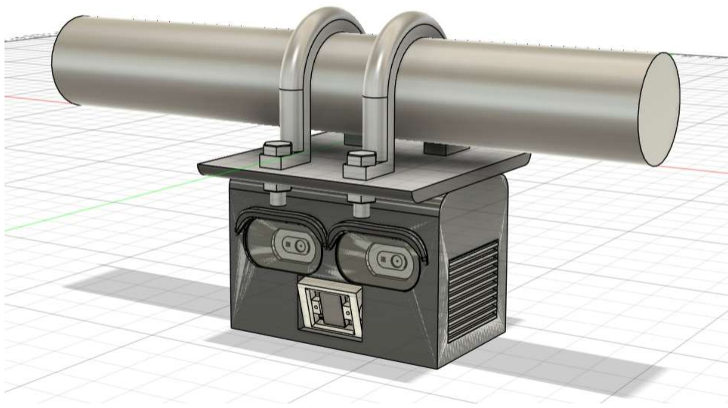
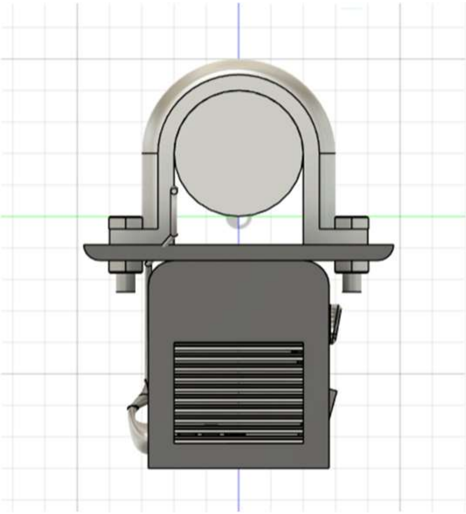


Fig.: CAD Model

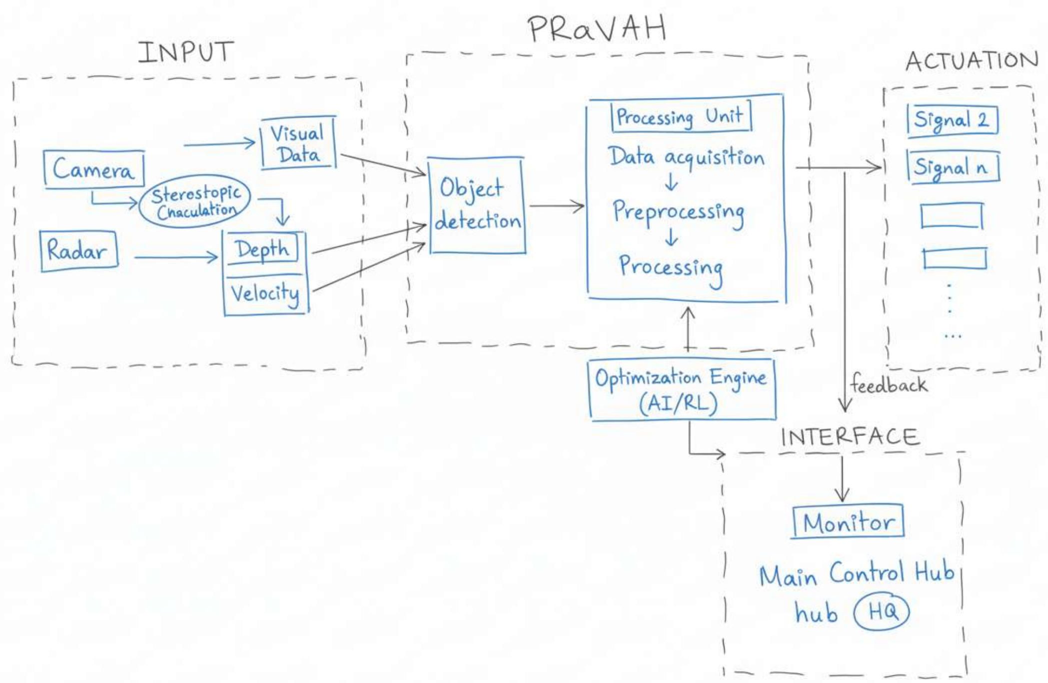


System Architecture

Our Workflow Explained:

- **Sensor Input:** The process begins with our dual cameras and radar collecting raw data - visual feeds, depth maps, and velocity measurements from the traffic on the ground.
- **Fusion & Processing:** This raw data is fused and pre-processed inside the device. Our algorithms perform **real-time object detection and classification**, identifying vehicles, pedestrians, and other objects.
- **Core Analysis:** The system then analyzes this information to understand the complete traffic scenario, calculating key metrics like **vehicle queue length, traffic density, and flow rate**.
- **Intelligent Actuation:** Finally, our **Reinforcement Learning (RL) Optimization Engine** uses this analysis to make a decision. It sends commands to dynamically change traffic signal timings and transmits crucial alerts to the traffic headquarters dashboard.

Work-Flow


INPUT
PRaVAH
ACTUATION
Camera
Visual Data
Sterostopic Chaculation
Radar
Depth
Velocity
Object detection
Processing Unit
Data acquisition
Preprocessing
Processing
Signal 2
Signal n
...
feedback
Optimization Engine (AI/RL)
INTERFACE
Monitor
Main Control Hub
hub HQ

Additional Work Yet To Be Undertaken

What we aim to demonstrate in the next 4 months

We plan to enhance and validate the PRaVAH system through the following developments:

- **Scalability Testing:** Deploy the system across multiple intersections to demonstrate real-time synchronization and traffic optimization.
- **AI-Driven Predictive Analytics:** Integrate machine learning models for long-term traffic trend forecasting and early detection of congestion or hazards.
- **Enhanced Sensor Fusion:** Incorporate additional sensors such as **LiDAR** and **audio sensors** for more accurate environment mapping and object detection.
- **V2I Integration:** Begin pilot integration with **vehicle-to infrastructure(V2I)** communication systems to enable smarter vehicle coordination and emergency routing.
- **Performance Validation:** Evaluate system robustness, latency, and ease of use under different weather and traffic conditions.

What we intend to do after the competition

Post-competition, we aim to:

- **Scale Deployment:** Collaborate with **municipal bodies and Smart City Missions** for large-scale city-wide implementations.
- **Expand Use Cases:** Adapt PRaVAH for diverse environments including **highways, rural intersections, and event-based traffic management.**
- **Commercial Readiness:** Develop a cost-efficient, modular hardware kit and cloud dashboard for easy integration into existing traffic infrastructure.
- **Long-Term Vision:** Build a **data-driven urban mobility platform** that connects traffic management, emergency services, and public transport under one ecosystem.

Challenges Faced

Challenges & Risks for PRaVAH Implementation

- **Hardware Integration:** Difficulty in seamlessly combining multiple sensors (camera, radar, IoT units) into existing infrastructure.
- **Data Privacy & Security:** Managing and protecting large volumes of real-time video and traffic data.
- **Scalability Issues:** Ensuring consistent performance across thousands of intersections.
- **Environmental Limitations:** Harsh weather (rain, fog, dust) may affect sensor accuracy.
- **Adoption Barriers:** Municipal bureaucracy and slow decision-making for deploying new tech.
- **Maintenance & Cost:** Regular calibration and upkeep of sensors to ensure reliability.
- **Connectivity:** Dependence on stable network infrastructure for cloud and IoT data transfer.

Current Problems in the Domain

- Outdated **fixed-timer traffic lights** unable to respond to live traffic fluctuations.
- **Uncoordinated traffic management** across intersections leads to long queues and bottlenecks.
- **Delayed emergency response** due to lack of vehicle prioritization mechanisms.
- **Poor data availability** - no unified system for collecting, analyzing, and acting on real-time traffic information.
- Frequent **road disruptions** (accidents, rallies, waterlogging, road repairs) remain **unmanaged in real-time**.

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